

Information Gathering & Reconnaissance

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Course: Ethical Hacking Internship

Task: Information Gathering & Reconnaissance

Target Website: Savitribai Phule Pune University (SPPU)

Website: <https://www.unipune.ac.in>

1. Objective

The objective of this task was to understand the reconnaissance phase of ethical hacking by collecting publicly available information about a target website. Passive reconnaissance techniques were used to gather domain information, DNS records, website technologies, HTTP security headers, and robots.txt/sitemap details without interacting with or exploiting the target system.

2. Target Information

Website Name:

Savitribai Phule Pune University (SPPU)

Website URL:

<https://www.unipune.ac.in>

Reason for Selection:

Savitribai Phule Pune University was selected because it is a publicly accessible educational institution. The website provides sufficient publicly available information for practicing passive reconnaissance while ensuring that no unauthorized interaction or exploitation of the system is performed.

3. WHOIS Lookup

Tool Used

- ICANN WHOIS Lookup / WHO.IS

Observation

WHOIS lookup provides ownership and registration information of the domain, including the registrar, registration dates, domain status, and name servers. This information helps identify the administrative details of the target domain.

4. DNS Enumeration

Tool Used

DNSChecker

DNS Records

Record	Description
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A Record	Maps the domain name to its IP address.
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MX Record	Identifies the mail server responsible for receiving emails.
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NS Record	Lists the authoritative name servers for the domain.
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TXT Record	Contains additional information such as verification and security-related records.
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Observation

DNS enumeration helps identify important DNS records that reveal how the website is hosted, how email services are configured, and how domain name resolution is managed.

5. Website Technology Identification

Tool Used

BuiltWith

Technologies Identified

Technology Category	Identified Technology
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Web Server	Microsoft IIS 10
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Framework	ASP.NET
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Website Language	English (Inferred)
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Content Delivery Network	BootstrapCDN, jsDelivr, AJAX Libraries API, jQuery CDN
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JavaScript Libraries	jQuery (v1.12.4, v3.5.1, v3.6.0)
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JavaScript Components	Popper.js, Intersection Observer
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Social Media Integration	Facebook SDK
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Other Technology	Shockwave Flash Embed
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Observation

The technology stack indicates that the university website is hosted on Microsoft IIS and developed using ASP.NET. Multiple CDNs are used to improve content delivery, while JavaScript libraries such as jQuery and Popper.js provide client-side functionality.

6. HTTP Security Headers

Tool Used

SecurityHeaders.com

Security Header	Present Purpose	
Content-Security-Policy (CSP)	No	Protects against Cross-Site Scripting (XSS).
X-Frame-Options	No	Prevents clickjacking attacks.
X-Content-Type-Options	No	Prevents MIME-type sniffing.
Strict-Transport-Security (HSTS)	No	Forces secure HTTPS connections.
Referrer-Policy	No	Controls referrer information shared with other websites.

Observation

The analysis revealed that several recommended HTTP security headers were missing. Implementing these headers would improve protection against common web-based attacks and strengthen the overall security posture of the website.

7. Robots.txt & Sitemap Analysis

Robots.txt

Available: Yes

The robots.txt file specifies directories that should not be crawled by search engines, including /admin/, /private/, /backup/, /scripts/, /cgi-bin/, and /tmp/. It also provides the location of the XML sitemap.

Sitemap.xml

Available: Yes

The sitemap contains a structured list of important URLs related to admissions, academics, research, scholarships, university departments, finance, and other services. This helps search engines efficiently discover and index the website's content.

8. Overall Observations

The reconnaissance phase demonstrated how a significant amount of information about a website can be gathered using only publicly available sources. WHOIS records revealed domain registration details, DNS enumeration identified domain configuration, technology identification exposed the website's software stack, and HTTP header analysis highlighted areas where security could be improved. The robots.txt and sitemap.xml files provided valuable insight into the website's structure and publicly accessible resources. No active interaction with the target system was performed during this assessment.

9. Conclusion

The reconnaissance phase is one of the most important stages of ethical hacking because it helps security professionals understand the target before conducting any security assessment. During this task, passive information gathering techniques were used to collect publicly available information about the Savitribai Phule Pune University website. Various online tools were used to examine domain registration details, DNS records, web technologies, security headers, robots.txt, and sitemap.xml. The activity demonstrated how much information can be obtained without directly interacting with the target system. The analysis also identified several missing HTTP security headers that could be implemented to improve the website's security. This exercise enhanced my understanding of ethical information gathering techniques and reinforced the importance of conducting reconnaissance responsibly and within legal and ethical boundaries.